

BAPURAM VENKATESWAR REDDY

AI Engineer — Generative AI, RAG & Agentic Systems

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EDUCATION

National Institute of Technology Warangal (NIT Warangal) <i>Integrated M.Sc. (B.S. + M.S.) in Mathematics</i> Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science, Probability & Statistics, Linear Algebra, Optimization Theory, Graph Theory, Algorithms	Warangal, India 2021 – 2026
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EXPERIENCE

Devseccops AI (Sec and DevOps AI Pvt. Ltd.) <i>Full-Stack AI Engineer Intern</i> - Built enterprise Generative AI and RAG applications using Python, FastAPI, LangChain, LangGraph, FAISS, Chroma, and AWS Bedrock. - Reduced LLM inference costs by 35% through model optimization, token minimization, and prompt caching strategies. - Designed agentic AI workflows with state, context, tools, and structured execution; reduced unnecessary LLM calls by 25%. - Built multimodal pipelines for text, documents, speech, images, and video using Amazon Nova, Gemini, Whisper Large v3, Ray 2, and FFmpeg. - Achieved 2x higher media-processing throughput through parallelized worker queues and async execution pipelines. - Reduced manual media-processing effort by 50% and automated video composition transitions, reducing visible flickering by 40%. - Developed deployable AI backend services containerized with Docker and deployed on Amazon EC2 and Amazon S3. - Contributed to quantitative forecasting workflows using XGBoost and Random Forest, achieving 15% lower forecasting error.	Jul 2026 – Present
WorldQuant BRAIN <i>Research Consultant (Part-time Remote)</i> - Designed and backtested 20+ quantitative alpha strategies across global equity datasets with multi-regime risk analysis. - Applied time-series analysis, statistical validation, and drawdown limits, achieving Sharpe ratios consistently above 2.0.	May 2025 – Jun 2026

FEATURED PROJECTS

Multi-Agent Claim Processing AI System <i>[LangGraph, FastAPI, OCR, LLMs, Docker, Python]</i> Engineered an agentic document intelligence workflow transforming unstructured insurance claim PDFs into structured JSON with schema validation, intelligent routing, and automated claim status updates.
Financial AI Assistant (RAG) <i>[RAG, LangChain, FAISS, FastAPI, Semantic Search]</i> Developed a sub-second retrieval-augmented generation platform for complex financial reports using dense vector indexing, contextual chunking, and source attribution.
ADGM-Compliant Corporate Agent with Document Intelligence <i>[LangChain, FastAPI, Python, Document Intelligence, OCR]</i> Built a regulatory compliance agent parsing statutory filings and evaluating legal submissions against ADGM guidelines with traceable audit logs.
Automatic Number Plate Recognition System <i>[YOLOv8, OpenCV, OCR, PyTorch, Python]</i> Trained and benchmarked a near-real-time license plate detection and character extraction pipeline evaluated on 1,000+ annotated traffic frames across adverse conditions.
Logistics AI Tracker <i>[React, FastAPI, LangChain, RAG, Docker, GitHub Actions]</i> Created a full-stack supply chain tracking platform featuring a natural-language copilot for shipment incident queries with containerized CI/CD deployment.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, C++
Generative AI & Agents: LLMs, RAG, LangChain, LangGraph, Agentic Workflows, Multi-Agent Orchestration, Prompt Engineering, Tool Calling
Models & Platforms: AWS Bedrock, Amazon Nova, Gemini, Hugging Face, Whisper Large v3, OpenAI APIs, Anthropic APIs, Ray 2
Retrieval & Vector Search: FAISS, Chroma, Vector Databases, Dense Embeddings, Semantic Search, HF Embeddings
Backend & Cloud: FastAPI, Node.js, REST APIs, Async Processing, Docker, Amazon EC2, Amazon S3, GitHub Actions, CI/CD, Linux
Machine Learning: Scikit-learn, PyTorch, XGBoost, Random Forest, Time-Series Forecasting, Statistical Validation, Model Evaluation
Multimodal AI: Speech-to-Text, OCR, Video Processing, FFmpeg, Text-to-Video, Video Generation, Structured Data Parsing

HONORS & LEADERSHIP

- International Quant Championship 2025: Global Rank 156 in Stage 1, conducted by WorldQuant.
- Flipkart GRID 6.0: National Semi-Finalist (Top 1% selected among 10,000+ engineering teams in India).
- All India Inter-NIT Yoga Championship: Second Runner Up (National Podium across all NITs).
- Additional Secretary, Mathematical Society, NIT Warangal (2022 – 2025).
- Event Coordinator, SpringSpree Cultural Festival, NIT Warangal (2021 – 2025).